

# Haocheng Liu

☎ (+1) 412-245-6253 | ✉ lbtinglb@gmail.com | 🌐 haochengLIU | 🌐 haocheng-liu

## Skills

**Programming** C/C++20, CMake, Python, make, Git, Bash, Java  
**DevOps** Linux, Multi threading, Meta programming, Google Cloud, GDB, Vagrant, Docker  
**Open source** Apache Arrow, VTK, ParaView, SMTK, CMB, ADIOS2, Spack, CMake, Rasberry Pi

## Work Experience

### Two Sigma Investment LP

New York, NY

SENIOR SOFTWARE ENGINEER IN MARKET DATA PLATFORM ENGINEERING

Dec. 2019 - Now

#### • Tech Lead - Architected a Performant and Scalable Market Data Platform for Research and Analytics

- Leading a team of 2 engineers on a multi-year initiative to develop an open-standard, cloud-native platform handling 200+ TB of data daily and serving 200+ diverse users, hosting several petaBytes of ever-growing data.
- Architected a high performance data processing framework in C++ with Apache Arrow and multi-threading, serving users with varying endpoint and latency requirements, handling 2000+ market data feed variations in a hybrid machine environment, reducing in compute and storage cost by 50%.
- Designed a complete and sustainable integration with proprietary modeling platform, BigQuery via Terraform and Apache Iceberg via external commit, coordinating efforts across multiple cross-functional engineering and research teams.
- Driving the user onboarding across organizations, working with partner teams to resolve tens of millions of PnL discrepancies, launching new fast trading models across regions, and saving compute costs in key pipelines by up to 90%.
- Debugged abnormal memory usage and S3 authentication issues, Optimized query performance, achieving >5x with ad-hoc intraday queries and >100x with large queries spanning months of data and contributing features and patches to Apache Arrow.

#### • Tech Lead - Onboarded a Conversational Intraday Microstructure Market Data Analytics System

- Independently evaluated offerings from eight vendors to close the gap on serving streaming, micro market data structure feature generation for low latency trading and data governance for the firm.
- Designed evaluation criteria, conducted rounds of internal and external interviews, performed proof of concepts with vendors, cloud and security teams and wrote a detailed report which guided the senior management to onboard the selected vendor.
- Onboarding the chosen vendor offering to existing platform supporting all US depth of book feeds and future exchanges.

#### • Maintained and Enhanced High-Throughput, Low-Latency Market Data Platform

- Led a team of 3 engineers to expand company's trading portfolio to Asia, onboarding 6 direct feeds and backfill decades of research data, launching new trading tactics.
- Developed a generic C++ framework and corresponding Python bindings to historical all market data in a proprietary format, leveraging std::variant, templates and Pybind11, unlocking new research opportunities for modelers.
- Implemented real time, low latency market data feeds in C/C++ from major stock exchanges all over the world, leveraging various optimization techniques in C++ and OS.
- Redesigned fault-tolerant tickerplant health reporting, contributing to the new market data monitoring and metrics system on top of Prometheus and Grafana.

#### • Community and Culture

- Modernized the company C++ development toolchain by migrating a bespoke build system to industry open standard toolchain CMake.
- led the GCC 13 compiler upgrade for the firm, upgrading 80+ codebases, certifying 500 codebases across multiple orgs.
- Presented work at organization townhall, company wide technique sharing session and teaching classes on "Market data for research".

### Kitware Inc.

Clifton Park, NY

R&D ENGINEER

Jun. 2016 - Dec. 2019

#### • Scientific multi-threading algorithms for Massively Threaded Architectures toolkit(VTK-m)

- Simplified and optimized the infrastructure which saved runtime overheads by CRTP pattern and generic programming. It reduced the binary size by 80% and improved 40% of runtime performance
- Refactored the code base with C++11 features which simplified meta programming, improving runtime overhead as well as adding a reliable random number generation and prepare for C++14 migration

#### • Simulation Modeling ToolKit and Computational Model Builder framework

- Architected a simulation tool kit which provided a front end UI to design and visualize complicated model via Qt and GPU glyphing, adding C++11 python binding support and ability to launch parallel meshing jobs via a thread pool with commercial meshing libraries

## Education

### Carnegie Mellon University

Pittsburgh, PA

M.S. IN MECHANICAL ENGINEERING 3.7/4.0

Aug. 2014 - May. 2016

### Shanghai Jiao Tong University

Shanghai, China

B.S. IN MARINE ENGINEERING 3.7/4.0

Sep. 2010 - June. 2014